



2 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{4}{9}(x+1) & 0 \leq x \leq 1, \\ (x-2)^2 & 1 < x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the cumulative distribution function of X . [3]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(b) Find the exact value of the median of X . [2]

[illegible]



The random variable Y is defined by $Y = \sqrt{X}$.

- (c) Find the cumulative distribution function of Y . [3]

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- (d) Determine whether the median of Y is greater than, or less than, the median of X . [2]

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